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ASSOCIATE SERIES · A01 · SECTION 1 PLATFORM (6%) QUICK REFERENCE

# Architecture · Unity Catalog · Delta Lake

Section 1: core components (architecture, Delta Lake, Unity Catalog) and compute choices.
Free Edition: serverless only, Unity Catalog, one 2X-Small Structured Query Language (SQL) warehouse.
Fictional rows only.

## 1. Two planes

```
control plane  -> Databricks-managed services + web app (Databricks account)
compute plane  -> where data is processed
  serverless   -> serverless compute plane in the Databricks account
  classic      -> virtual machines (VMs) in YOUR cloud account
```

Account: identity, workspaces, metastores, usage. Workspace: notebooks, jobs, queries, dashboards.

## 2. Unity Catalog three-level namespace

```
metastore -> catalog -> schema -> table | view | volume | function | model
SELECT * FROM catalog_name.schema_name.table_name;
SELECT current_catalog(), current_schema();
```

One metastore can attach to several workspaces in the same region. Unity Catalog governs access, lineage, audit.

## 3. Managed Delta table (no LOCATION = managed)

```
CREATE SCHEMA workspace.bt_platform_xxxxxxxxxxxx;
CREATE TABLE workspace.bt_platform_xxxxxxxxxxxx.study_plan (
  lesson_id STRING, section STRING, topic STRING, weight_pct INT);
INSERT INTO ... VALUES ('A01','S1','Platform',6), ('A02','S2','Ingestion',21), ('A07','S3','Transformation',22);
```

Delta is the default table format on Databricks. Unity Catalog manages the files and their lifecycle.

## 4. Inspect it

```
DESCRIBE DETAIL study_plan;          -- format = delta, numFiles, sizeInBytes, createdAt
DESCRIBE TABLE EXTENDED study_plan;  -- Type = MANAGED
UPDATE study_plan SET topic = 'Transformation and modeling' WHERE lesson_id = 'A07';
DESCRIBE HISTORY study_plan;         -- 0 CREATE TABLE · 1 WRITE · 2 UPDATE
```

Delta = Parquet data files + a transaction log: atomicity, consistency, isolation, durability (ACID), schema enforcement, versions, time travel.

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# Compute services and cost models

## 5. Compute options

```
serverless notebooks   interactive Python/SQL, no config      (Free Edition)
SQL warehouse          SQL, business intelligence (BI), dashboards: serverless | pro | classic
                       Free Edition: ONE warehouse, 2X-Small
all-purpose (classic)  interactive, custom size/runtime, auto termination
jobs compute (classic) created per job run, ends with the run
instance pools         idle instances ready -> faster start
serverless jobs/pipelines  Lakeflow Jobs / pipelines, no infra
```

Free Edition does not support custom compute: all-purpose, jobs compute and pools are concepts here.

## 6. Cost models

```
DBU = Databricks Unit of processing capability, billed per second of use
classic    : DBUs to Databricks + virtual machine (VM) cost from your cloud provider
serverless : compute in the Databricks account, covered by the DBU price
jobs compute DBU rate  <  all-purpose DBU rate
pools      : no DBUs while idle, cloud provider still bills instances
```

Scheduled production work belongs on jobs compute or serverless jobs, not an interactive cluster.

## 7. Free Edition limits

Serverless only · one 2X-Small SQL warehouse · max 5 concurrent job tasks · one active pipeline per type · restricted outbound internet · no account console · daily fair-usage quota (compute pauses, data kept).

## Practice challenge

DELETE rows WHERE weight_pct < 10, then DESCRIBE HISTORY. Answer: 2 rows remain (A02, A07); newest entry is DELETE; read A01 with VERSION AS OF the version before the delete (until VACUUM removes old files).

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